No 487 MCC

GMCC FAMILIARIZATION

(ENGG OFFICERS)

ENGINEERING BRANCH

Mission

"Maintain all Weapon Systems at the Highest state of Operational Readiness with safety"

Vision

"Self reliance through HRD, Research and Indigenization"

Core Values

"Technical Discipline, Quality, Resilience"

"A reference booklet for MCC's Engg Officers"

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PREFACE

1. Technical proficiency is achieved through skillful application of professional knowledge, conduct of scheduled / unscheduled maintenances 'By-The-Book' and a culture of safety and work ethics. It is the prime responsibility of engineering officers to not only educate the technical manpower on the importance of these three tenets but to also lead the way for effective implementation at field level. Over a period of time, arising trend has been observed in Engg Officers to work through an 'off-hand' approach and rely on feedback from technicians. This approach not only results in unsafe maintenance practices but also undermines the 'leadership' role of Engg officers in technical domain.

2. With an aim to serve as first-hand Knowledge to newly posted-IN Engg Officers in MCC, an effort was initiated for compilation of necessary tasks under the lights of instructions of Headquarters Air Defence Command, PAF. Compiling tasks was successfully completed.

3. This booklet consists of information related to Organizational layout of MCC and its complete role and tasks in engineering domain.

4. This booklet will not only help in better understanding the role and task of MCC but also it will benefit Engg Officers in ensuring serviceability of equipment along with human source safety through necessary guidelines / route-map.

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INTRODUCTION

1. OVERVIEW

MCC Units in static or mobile versions (GMCC/MMCC) are the control and reporting centres of AD Networks. The GMCC receives data from multiple heterogeneous sensors and after filtration generates a composite air picture which is up told to SMCC. Each GMCC has low level radars as its integral part and deployed as per requirement.

1.1 Organizational Hierarchy

The command layout of MCC is as follows: -

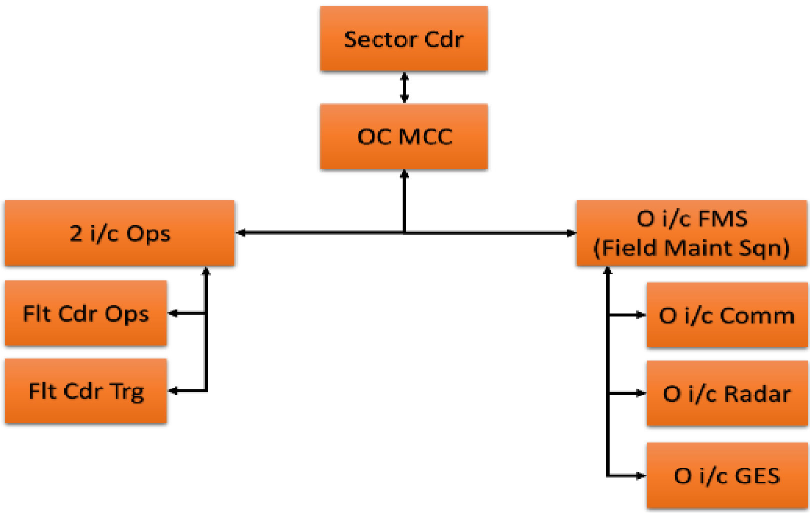

Fig 1.1: Organizational Hierarchy

FIELD MAINTENANCE SQUADRON
(FMS)

2. O I/C FIELD MAINTENANCE SQUADRON (FMS)

He is responsible to the Officer Commanding Wing for the following: -

2.1 Role and Tasks

(a) Control and direction of supply and maintenance organization of the Unit.

(b) Maintenance of technical publications and implementation of technical orders issued by AHQ.

(c) Liaison with concerned authorities for maintenance of landline communications.

(d) Maintenance of cypher security and communication orders.

(e) Review record of maintenance accomplishment and updation of equipment inventories and UAL.

(f) Maintenance, review and updating of the bench stock and pre-issue holding of spares.

(g) Indoctrination and encouraging ground safety programs and cost consciousness in maintenance accomplished by the personnel.

(h) During operational deployments of the Unit, he is to work out conveyance requirements including MT vehicles or air transport and arrange procurement. Subsequently, he is to ensure appropriate supervision during dismantling and installation of all the operational equipment.

2.2 Policies / Advisories Issued by HQ ADC, PAF

For continuous improvement in procedures being implemented for safeguard of men and material, O i/c FMS is to ensure implementation / adherence of advisories issued for undermentioned aspects related to Air Defence Setup vide ***Advisories for Technical Personnel Vol-I***: -

(a) MPDR

(b) Radar General

(c) Defcomm

(a) RT Equipment

(b) PAFComm

(c) Field Communication

(d) Weather

(j) Communication General

(k) Generators

(l) Air Conditioners

(m) MT Ops

(n) Automation

(p) Auto-comm

(q) UPS/MUPS

(r) Quality Assurance

(s) Safety and Security

(t) Travel Advisories

GMCC & COMM FLT

3. O I/C COMMUNICATION / GMCC

He is responsible to the O i/c FMS for the following: -

3.1 Role and Tasks

(a) Look after all the communications related problems of GMCC and radar sites.

(b) Liaise with PTCL / SATCOM Hubs / relay sites for effective reporting of faults.

(c) Ensure the availability of spare support of all communication equipment.

(d) Ensure prompt action for rectification of communication & networks fault in shortest time.

(e) Inform O i/c FMS about any un-serviceability or damage to the equipment at Wing Headquarters / deployed at sites.

(f) Keep track of all maintenance activities at GMCC, field sites and storage.

(g) As O i/c Comm & GMCC, he must have knowledge of technical and correspondence/ documentation aspects. Following are the important areas for this domain.

3.2 Specialist Directorates

Concerned specialist directorates at AHQ:

(a) Dte of Network Command (NWC)

(b) Dte. of Network Bases (NWB)

(c) Dte. of AE&S (Air Field Elects and Simulator)

(d) Dte of ADAS (Air Defence Auto Data System)

(e) Dte of AVCS

3.3 Types of Medias. Generally, Connectivity of different radar sites with GMCC is carried out utilizing **four types** of medias, as shown in following diagrams:-

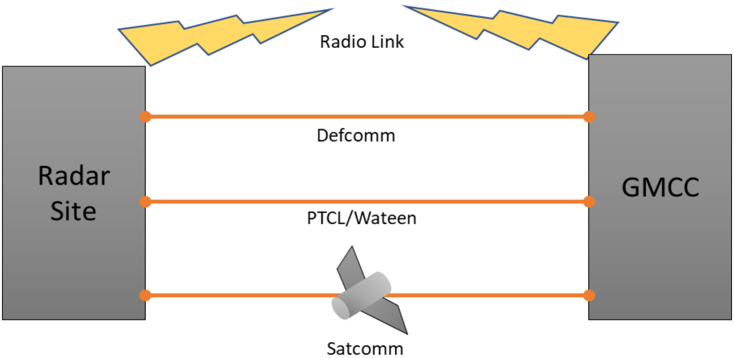

Fig 3.1: Four Types of Comm Medias

(a) **Radio Link.** A radio link is established from radar site to GMCC for connectivity. This media is utilized only if line of sight (LoS) between radar and GMCC is clear or at least up to 2F (2nd Fresnel).

(b) **LoS Softwares.** Line of sight for connectivity is determined using following Softwares:

(a) Radio Mobile (open source)

(b) EDX (USB dongle)

(c) Linkui (online)

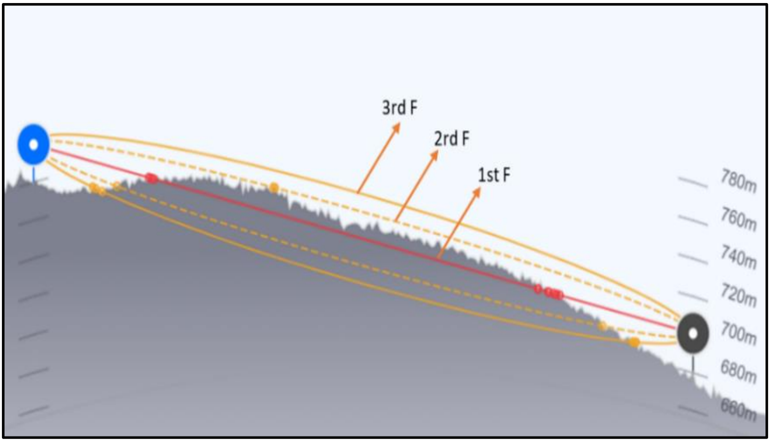

Fig 3.2: Three Fresnel zones of LOS

(c) **RL Equipment.** Selection of radio system for connectivity is carried out as per operational requirements. Four types of RL systems are being used. Specifications of RL systems are tabulated below: -

S no	Type	Frequency (Mhz)	Range (nm)	Data Rate	Encryption	Output
a	FM-200	610-960	45-60	TDM 256,512,1024 (kb/s)	No Encr (BEU)	Channels
b	RL-432	1350-1850	45-60	TDM 256, 512, 1024,2048 (kb/s)	No Encr (BEU)	Channels
c	RL-532	1350-2690	45-60	TDM 256, 512, 1024,2048 (kb/s)	No Encr (BEU)	Channels
d	HCLOS	4400-5875	90-110	476 mb/s	AES-128 AES-256	Ethernet form

(d) **Defcomm Media.** Ring connected optical fiber cable is utilized for combined Defence communication. Radar data from nearest Defcomm center can be fed over OFC utilizing E1s for further access at GMCC. Configuration of E1 can be carried out by approaching project management board (PMB) at GHQ. RMC SMG ring connectivity is with Mastung, Zhob, and Sibi.

(e) **Satcomm Media.** Connectivity of radar with GMCC may be carried out utilizing satellite communication. Following Satcomm equipment for connectivity with satellite at site is used:

(i) **COTP.** Communication On The Pause (COTP) is a vehicle based satellite transceiver. Primary Hub (GHQ) and Secondary Hub (PAF, Malir) is responsible for E1 configuration of COTP.

(ii) **Man-Pack.** This type of Satcomm equipment can be carried as back pack. Primary Hub (GHQ) and Secondary Hub (Malir, PAF) is responsible for maintenance and E1 configuration of Man-pack.

(iii) **Fly-Away Kit.** This type of Satcomm media is similar to Man-pack, however maintenance and E1 configuration is carried out by 221 and 223 TMS (Nurkhan, Faisal Base).

(f) **External media (PTCL, Wateen).** Connectivity of GMCC with radar sites may be carried out utilizing PTCL or Wateen as primary or secondary medias. E1 may be hired by approaching Dte of Network Command (NWC) through Headquarters Air Defence Command.

3.4 Architecture of GMCC and Vision System

A GMCC has Ops Room and CER. Ops room houses 12 control stations and communication equipment room (CER) houses radio, data processing and voice communication equipment. Following term are commonly used in GMCC:

(a) **Radar Data**. It is the raw data available in different forms (Azimuth, range, IFF) to extract required information.

(b) **Voice.** It includes UHF channel and Controller's communication of all radars.

(c) **Plots**. The unprocessed data or raw data displayed on console.

(d) **Track let.** The information regarding Speed, Heading and Quality of the object is called Track let.

(e) **Track.** The Tracks are developed by processing the Plots to indicate the Range, Azimuth, Bearing and Speed of moving Targets.

3.5 Data

Radar data is fed to GMCC from four resources i.e High level radars, Low level radars, COP cloud and NGN. The data is further processed in undermentioned sequence:

(a) **Cisco Router.** Router collects the data from different resources and direct them to LAN switch**.**

(b) **FSK Modem**. The data from low level radars is received in analog form (Frequency Shift Keying). This data is further fed to FSK modem which converts it to RS232 form.

(c) **Moxa Server.** It collects data in form of RS232, converts in Ethernet form directs it to LAN Switch.

(d) **DCG.** Data Communication Gateway receives all types of radar data packets in digital form and filters out the required data and sends it to MST.

(e) **MST.** Multi Sensor Tracker assigns track ID and track numbers to respective radar data. Multi tracker fusion module fuses same tracks received from different radars on the bases of absolute tracking (selecting superior radar).

(f) **CIS**. Central Information Server stores record and log of data. It has storage capacity of 8 TB and 32 RAM.

(g) **GDB.** Gateway Data Broadcast send and receive data tracks to COP cloud. All GDB servers are mesh connected.

(h) **ADSB-R.** Automatic Dependent Surveillance Broadcast (receiver) is passive information from airliners i.e. location, altitude, speed etc.

(j) **KVM Switch.** Keyboard Visual Mouse switches displays of servers over a single display.

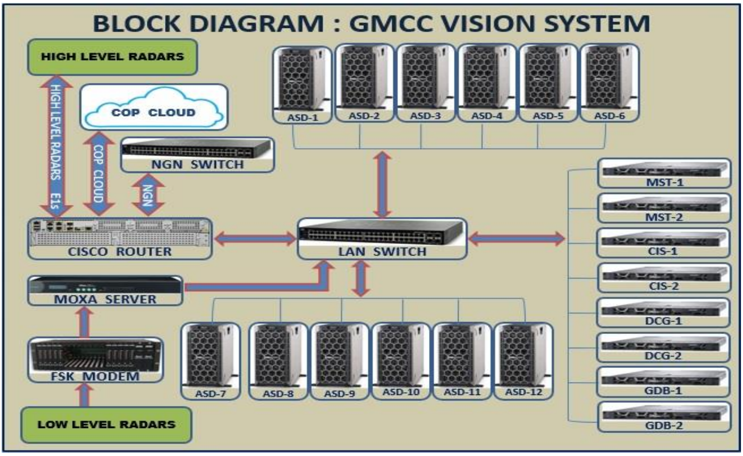

Fig 3.3: GMCC Vision System Layout

3.6 Voice

All types of voice communication in GMCC comprising Pafcomm, Defcomm, SHL, Field phone and UHFs are received in form of E1s. Which is fed on FMX/ Dynaflex where it is further segregated into channels. Block diagram of voice circuit is shown in figure below:

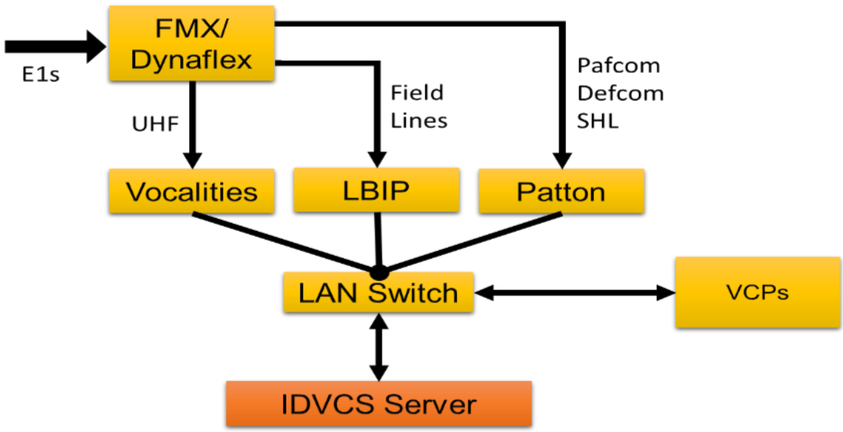

Fig 3.4: Voice Data Block Diagram

3.7 Hardware Equipment

(a) **FMX/ Dynaflex** *(Ref: communication Précis, 107 AED)*

Flexible multiplexer / Dynaflex merges multiple inputs (channels) to a single output (E1). Its integration capacity depends upon number of installed cards.

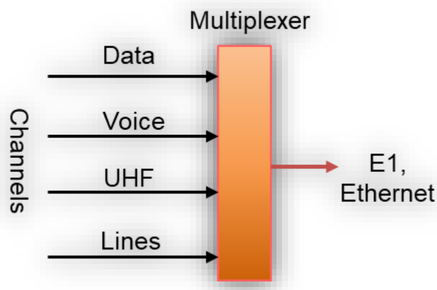

Fig 3.5: FMX/ Dynaflex

(b) **Optimux**

Opti-Multiplexer converts electrical signals to Optical signals. It has multiple input streams (ports) and

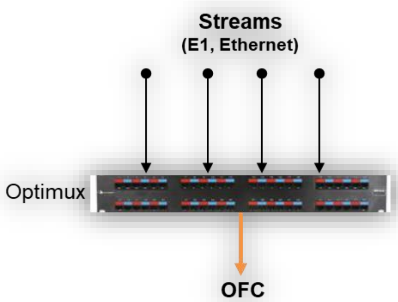

Fig 3.6: Optimux

single optical output. Optimux does not require configuration (Plug & Play).

(c) **OSN**

Optical Switching Network merges multiple streams (E1s) into a single optical stream which is further transported via Optical Fiber Cable (OFC). Its capacity depends upon the model (500, 1500, 3500) of OSN used. OSN is configured through software.

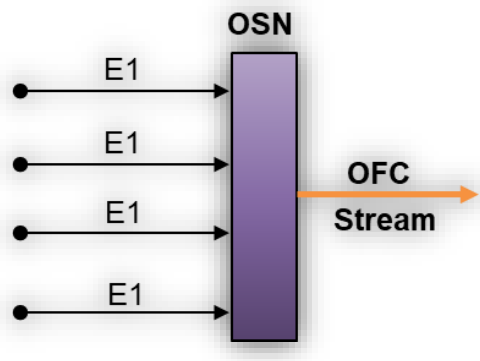

Fig 3.7: Working of OSN

(d) **M3SR** *(Ref: communication Précis, 107 AED)*

M3SR is one of the (M3) multiband, multimode and multirole transceivers radios. Airborne, Surface and tactical radios are three types of M3SR. M3SR operates in three modes; Plain, Comsec and Transec.

(e) **DTS**

Discipline Timing System synchronizes radio stations (airborne & surface) for communication. It has a built in GPS which extract time of the day.

3.8 GMCC Power layout

GMCC is operated on WAPDA and generator (in case of load shedding) powers. However, UPS is utilized for changeover of power from different sources. The power layout of GMCC is shown below:

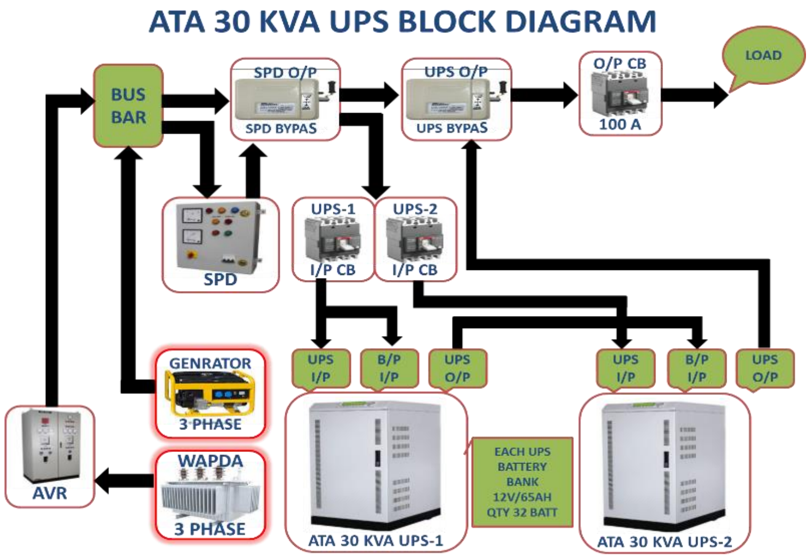

Fig 3.8: GMCC Power Layout

3.9 RCN (Report Control Number)

Report Control Number is a monthly /quarterly /six monthly or annual report of data pertaining to operational state /health /serviceability /performance of all ground Radars, C2 systems. RCN related to MPDR radars are given below:

S No	RCN Name	Addressee	Date (Monthly)
(a)	Monthly air staff presentation data Comm	WESSEC	1st
(b)	advisory on non-availability of AD data	WESSEC	2nd
(c)	Unserviceability of Air Defence connectivity due to media issue	WESSEC	2nd
(d)	Earthing value state COTP equipment	CMC Defcomm MLR	3rd
(e)	Monthly revised Comm equipment state	WESSEC	10th
(f)	Monthly verification of standby media (Fortnightly)	WESSEC	3rd
(g)	Judicious utilization of long-haul Data links	WESSEC	5th
(h)	PMI of Satcomm sites (PMB)	223 TMS	5th
(j)	Testing of Satcomm Network	CMC Defcomm MLR	25th
(k)	Monthly R & S M3sr radio state	WESSEC	5th
(l)	Verification of standby media	223 TMS	3rd
(m)	Shifting of load and E1 record OSN 500	223	5th

3.10 Publications and Comm Manuals

(a) Communication précis No 107 AED

(b) C2 Vision System No 118 SED

RADAR FLT

4. O I/C RADAR

O i/c Radar is responsible to O i/c FMS for the following duties:-

4.1 Role and tasks

(a) Handle all maintenance problems of Radar (MPDR).

(b) Maintain liaison with fault control sections and standby maintenance party and ensure their availability.

(c) Liaise with radars for effective reporting of faults.

(d) Ensure the availability of spare equipment for all ranges of equipment.

(e) Ensure prompt action for rectification of fault in shortest possible time.

(f) Ensure timely dispatch of equipment at sites if required.

(g) Inform O i/c FMS about any un-serviceability or damage to the equipment at Wing Headquarters / deployed radar sites.

(h) Keep track of all maintenance activities at Wing Headquarters and radar sites.

4.2 Specialist Directorates

(a) Dte of SRR (Short-Range Radars)

(a) Dte of LRR (Long-Range Radars)

(b) Dte of SAMS (Surface to Air Missile Systems)

(c) Dte of AES (Air Field Electronics and Simulator)

(d) Dte of ADAS (Air Defence Automation System)

4.3 Important Publications & Manuals

(a) AFM 355-2 (Manual of Air Defence)

(b) AFO 103-1 (Maintenance Management of AD Radars)

(c) AFO 103-2 (Safety Precautions: AD Radars)

4.4 MPDR (Mobile Pulse Doppler Radar)

The purpose of MPDR is to detect low flying aircraft and indicate the target information (range and azimuth) on the Plan Position Indicator (PPI) screen or to remote location for display. MPDR works in MTI mode (Moving Target Indicator), completely free of fixed target indication. Different types of MPDRs and their specifications are mentioned below:-

SPECIFICATION	MPDR-45	MPDR-60	MPDR-90
Mode of Operation	MTI		
Frequency range	1215 ~ 1365 MHz (in L band)		

Detection range (RCS of a target of 1m²)	1.5 ~ 45 Km	1.5~60 Km	1.8~92.5 Km
Detection altitude	14760 ft		20000 ft
Detection Probability	90 %	80 %	70 %
Target resolution (Range)	0.27 NM (500 m)	-	0.25 NM (460 m)
Target resolution(azimuth)	4°		
Location accuracy	±250m	-	±230m
Pulse duration	3.3 μ sec		
Average power (watts)	200	100 & 133	67 & 100
Peak power	20 Kw		
Input sensitivity (MDS)	-110 dBm	-115 dBm	-117 dBm
Time for frequency change	3 sec		
ECCM features	Constant False Alarm Rate		
	Amplitude versus azimuth		
	Pulse length discriminator		
	Inter pulse suppression		
	Frequency change		
	Constant false alarm rate		

4.5 Types of Maintenance

There are three types of maintenance activities including scheduled, preventive and corrective maintenances falling under the Radar Engineering domain.

SCHEDULED MAINTENANCE TIME : MPDR			
Daily	**Fortnightly**	**Four Monthly**	**Annual**
45 min	02 Hrs	02 Days	07 Days

4.6 RCN (Report Control Number)

RCN is a monthly /quarterly /six monthly or annual report of data pertaining to operational state /health /serviceability /performance of all ground Radars, C2 systems. RCN related to MPDR radars are given below:

S No	RCN Name	Addressee	Date (Monthly)
(a)	Air Staff Presentation Data	WESSEC	1st
(b)	Radar State (Through post guard mailing system)	WESSEC	2nd
(c)	Maintenance Up-keeping of Storage Radars	WESSEC	2nd
(d)	RCN-269 MPDR	SRR	3rd
(e)	Prompt Dispatch of RD Items	WESSEC	28th

4.7 Safety Precautions *(Ref. AFO 103-2)*

Personnel working on radar equipment are exposed to the following hazards: -

(a) RF radiation. Microwave radiation emitted from radar transmitter produce physiological damage to personnel.

(b) X-Ray radiation. Magnetrons, power klystrons, hydrogen thyratrons and HV rectifiers used in radar transmitters emit X-Rays during operation.

(c) Fire hazards due to RF radiation. Fire hazardous items (Fuel tanks, gasoline, POL, and ammunition stores) must not to be placed within 100ft from radar installation.

(d) High voltage. The RF and modulator assemblies contain points carrying lethal voltages of 50V DC/ 30V AC or greater and current of 9 mA or greater, which is potentially dangerous for human body.

GROUND ENGINEERING SECTION (GES)

5. O i/c GES (Ground Engineering Section)

O i/c GES is responsible for maintenance of MT vehicles and AD generators. He is responsible to O i/c FMS for the following duties:

5.1 Role and Tasks

(a) Ensure serviceability and availability of all MT, specialist vehicles and generators.

(b) Ensure the availability of spare equipment of all MT / Specialist vehicles and generators.

(c) Ensure prompt action for rectification of fault in shortest possible time.

(d) Keep a track of all the movements of MT vehicles.

(e) Regularly brief MTDs and CDs about safe driving techniques.

(f) Ensure all MTDs / CDs have valid driving license and TMAs.

(g) Keep track of all maintenance activities at WHQ and radar sites.

5.2 Specialist Directorates

(a) Dte of Short-Range Radars (SRR)

(b) Dte. of Ground Transport (Grd Tpt)

(c) Dte. of Network Command (NWC)

(d) Dte. of Medical Services (DMS)

5.3 Important Publications and Manuals

(a) AFO 77-17 (Organization of PAF Mech Transport)

(b) AFO 77-01 (Usage of Service Transport)

(c) AFO 77-09 (Safeguard of MT Vehicles)

(d) MTD Trade Manual

5.4 MT OPS Flt

The operational efficiency of the PAF is largely dependent upon its Mechanical Transport (MT). The proper functioning and efficient operation of mechanical transport is, therefore, of utmost importance to the service. Three types of vehicles are used in Air Defence:

(a) **Air Defence Specialist Vehicles.** ET, AT, DT RVT, Unimog, Prime Mover, COTP…

(b) **Grd Tpt Specialist Vehicles**. Ambulance, Fuel Bowser, Flat bed, Water Trailer…

(c) **Common User Vehicles.** Single cabin, Double cab, Trucks, Motorcycle…

5.5 Responsibilities of MT Staff

Duties and responsibilities of various NCOs i/c and drivers employed in the MT Squadron are given below. O i/c MT may allocate any additional duties which he considers essential for efficient running of MT Squadron:

(a) NCO i/c MT Detail.

(b) NCO i/c MT Records.

(c) NCO i/c MT Stores.

(d) NCO i/c MT Yard.

5.6 MT Forms: *(Ref: AFO 77-01)*

(a) **Form-658.** Application for MT vehicle for duty journey

(b) **Form-658 A**. MT routine run or detachment record

(c) **Form-658 E**. Authority of MT vehicle for duty journey

(d) **Form-793.** Application for MT Vehicle on payment

(e) **Form-656.** MT servicing form (DI, Defects, Maint)

(f) **Form-814 C**. Claim for over time of civilian drivers

(g) **Form-10044**. MT Vehicle engine oil book

(h) **Form-813.** Vehicle log book

(j) **Form-525.** Daily record of MT journeys

(k) **Form-748.** Inventory of equipment of MT Vehicle

(l) **Form-11076**. Driver log book

(m) **Form-11750.** MT Servicing and Defect Reporting

(p) **Form-446 A.** Traffic Accident Report

(q) **Form-1839, 1629 A&B.** MT Service license for Officers, Airmen & Civilian

5.7 MT RCNs

Monthly and quarterly RCN reports of MT Ops flt are given below:

S No	RCN Name	Addressee	Date
(a)	Ground Safety Occurrences	WESSEC	1st of each Month
(b)	Economy in MT Utilization /KM state	WESSEC	2nd of each Month
(c)	State of Ambulance	DMS	5th of each Month
(d)	MT management review	WESSEC	
(e)	Monthly return Poor KMPL	GRD TPT	
(f)	RCN-747 MT monthly operations	GRD TPT	
(g)	Driving Training on Simulator State	WESSEC	5th (Jan, Apr, Jul & Oct)
(h)	RCN-750	GRD TPT	
(j)	Excess MT mileage	WESSEC	
(k)	Optimization of MT Vehicles Fuel Consumption	GRD TPT	
(l)	SMT Move on F-793	GRD TPT	
(m)	Categorization of MTDs	WESSEC	7th (Jan, Apr, Jul & Oct)
(n)	Data of Specialist Vehicles	WESSEC	
(p)	Mandatory Holding of TMA by the Operators of Spl Veh	WESSEC	2nd (Jan, Apr, Jul & Oct)

5.8 D&M FLT (Diagnosis & Maintenance)

Scheduled maintenance plays an important role in ensuring a long service life in case of all types of MT vehicles and AD generators. D&M Flt is responsible for the maintenance (repair) of mechanical parts and engine assembly while Electric Flt is responsible for electrical parts and Alternator of the vehicle and generators.

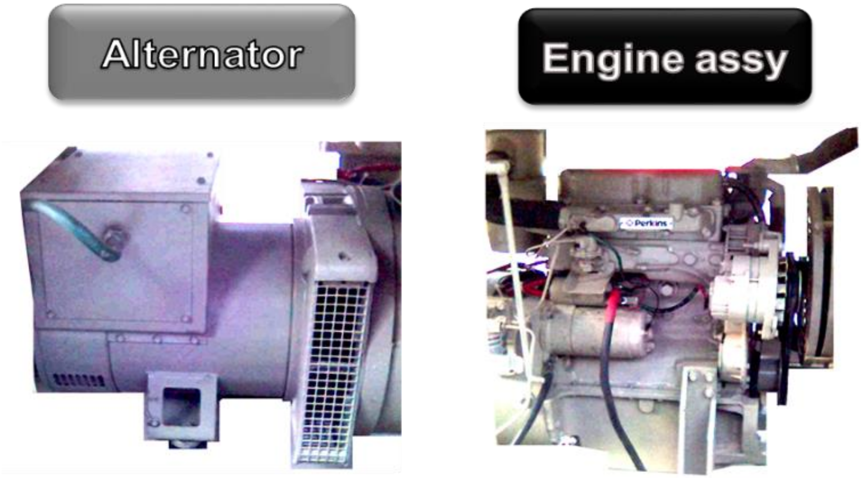

Fig 5.1: Main parts of Generator

5.9 EMI of AD Generators

Equipment maintenance inspection is to provide general instructions and safety precautions for the safety of personnel and AD generates. Given below are the scheduled inspections of AD generators:

Gen	Schedule Inspections	Remarks
15 / 20 KVA	30 Hrs	Initially for New /Overhauled Gen
	100 Hrs	Oil filter, Fuel Filter Engine oil etc.
	400 / 800 Hrs	-
	1000 Hrs	-
65 KVA	30 Hrs	Initially for New /Overhauled Gen
	250 hrs	Oil filter, Fuel Filter Engine oil etc
	500 hrs	-
	1000 or Six months	-
	2000 or Yearly	-

5.10 Life & Overhauling of AD Generators

The economical life of Powered Ground Equipment is laid down in terms of years to facilitate necessary forecasting of maintenance spares to plan the overhauls, and to assess the attrition rate. Following are the life and overhauling of AD generators is given below:

<table>
<tr><th rowspan="2">GEN TYPE</th><th colspan="2">Total Life</th><th colspan="3">OVERHAUL (Hrs)</th></tr>
<tr><th>Years</th><th>Hrs/Km</th><th>1st</th><th>2nd</th><th>3rd</th></tr>
<tr><td>10 KVA</td><td>15</td><td rowspan="3">25000</td><td rowspan="3">10000</td><td rowspan="3">7000</td><td rowspan="3">5000</td></tr>
<tr><td>15 KVA</td><td>15</td></tr>
<tr><td>20 KVA</td><td>15</td></tr>
<tr><td>60 KVA</td><td>20</td><td>30000</td><td>13000</td><td>8000</td><td>7000</td></tr>
</table>

5.11 Electric Flt

Electric Flt is responsible for electrical wiring of vehicles, generators, battery charging and maintenance of air-conditioning equipment. Scheduled maintenance and inspections of the AC are given below:

5.12 Daily/Weekly inspection

Daily and weekly inspections are to be carried out by the user. Applicable items of inspection and precautions are to be displayed near the AC.

5.13 Monthly, Quarterly & Annually

Inspection of ACs is to be carried out as per checklist given in AFO 78-21.

SOFTWARES

6. ESSENTIAL SOFTWARES (ALL DOMAINS)

Softwares used for different tasks in MCC unit are as follows: -

6.1 PAF OAS

Office Automation System (OAS) in Pakistan Air Force has revolutionized the work flow of physical file system. OAS is replacement of PAF Mail v2.0. The main objective of this system is to provide paperless office environment and minimum turnaround time for letters, tasks, DFAs, part cases and notifications.

Ref: *Firefox → OAS → User guide*

6.2 ADIMS

Air Defence Information Management software contain information of PAF radars, generators, C2 systems and Air Defence specialist vehicles. It represents the serviceability and current state of Air Defence equipment. User ADIM ID performa is processed through base IT center.

Ref: *Firefox → ADIMS → User profile → support*

6.3 PALMS

Pakistan Air Force Automated Logistics Management System (ALMS-ev/ PALMS) is a modern web server application. It contains five servers which are further divided into SDs (System Designators). User ID performa is processed via base log sqn and further authorized by Dte of LAS.

Ref: *Intranet → applications → Palms → user_manual*

6.4 VMS

Vehicle Management Software is utilized for management of service mechanical transports. It includes serviceability state and other information about SMTs. VMS ID is authorized to MT Ops Flt of base or unit.

6.5 SEP Manager

Training management of all SEP (5 level, 7 level) trainees is carried out in SEP manager software. Endorsement, evaluation and other tasks are performed in SEP manager. SEP manager ID is only authorized to OC unit, O i/c work center, O i/c training, Training coordinator and trainers of the unit.

***Ref**: SEP policy*

6.6 E-QAP

E-Quality Assurance Program is Internet explorer web-based software utilized for QA management of work centers. ID is authorized to O i/c sections and WO i/c QA.

LOG FLT

7. SUPPLY PROCEDURES (ALL DOMAINS)

Log Flt is responsible for timely processing different types of demands, fund tracking and coordination with base Log Sqn. Important factors and tasks of Log Flt are given below: -

7.1 Depot Demands (*Ref. AFO 67-74)*

This type of demands are initiated on PALMS by user and further processed from base Log Sqn (Jet store/Tech store).

7.2 LP Demands *(Ref. AFO 67-137)*

In case of non-availability of item at depot, concern depot provides an “AD” code. User may demand the required item from LP (local purchase) section through specific contractor.

7.3 LR Demands (*Ref. AFO 67-74)*

In case of unserviceability of item and non-availability of repair facility at depot, concern directorate authorizes for local repair of item through base LP section.

7.4 Billing of LP/LR

Funds are allocated for user by Dte of LAS (Logistic Administration Services) for each FY year (starting from 01 July CY - 30 June NY). Fund is utilized by user against LP/LR demands. Billing of same is carried out by Log Sqn and base account section.

7.5 ROCP Demands

Radar Out of Commission for Parts is urgent demand raised by user in case of critical requirement and processed by Log Sqn. Concern depot ship the ROCP demand on priority to Log Sqn and subsequently to user via fastest means i.e. service Aircraft/ Civil Aircraft / TCS. ROCP demands are initiated by Priority “03” and Urgency Justification Code (UJC) “AC” on PALMS.

7.6 AROCP Demands

Anticipated ROCP demand is initiated for U/S item of radar however, radar remains serviceability. AROCP demand is initiated by priority “03” and UJC “BC” on PALMS.

7.7 Pre-Issue/ Bench Stocks

Pre-issue/ Bench Stock items are authorized to user by concern Dte. Bench stock items are demanded after specific period however, demands of pre-issue items are carried out in case item is deficiency in stock.

7.8 UAL / Scaling *(Ref. AFO 67-59, 67-1)*

Units are authorized with different Unit Authorization List or Scaled items. UAL includes all the Tools, testers and ground equipment, however Scaling includes all other items approved by ministry of Defence. Moreover, addition/deletion in UAL/scaling may be carried out by approval of Ministry of Defence.

7.9 SOB / SIB

Store Outward Book is documentation action carried out by base traffic section in case item is to be dispatched outward. Moreover, Store Inward Book is documentation actions at receiver end. Form-604 is used for SOB actions.

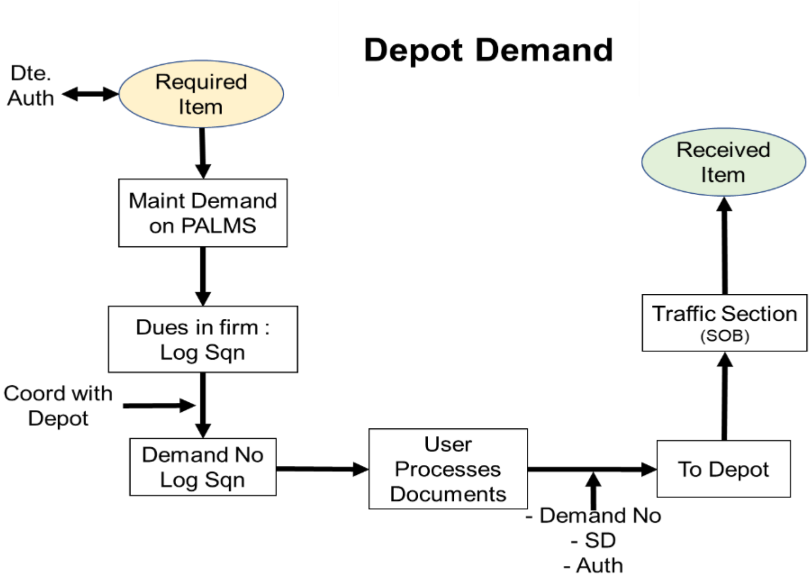

Fig 7.1: Flow Diagram of Processing Demands

GLOSSARY OF ABBREVIATIONS/ ACRONYMS

8. ABBREVIATIONS/ ACRONYMS

OJT	On the Job Training
GMCC	Generic Mission Control Center
MMCC	Mobile Mission Control Center
MPDR	Mobile Pulse Doppler Radar
ET	Equipment Truck
AT	Antenna Truck
CET	Communication Equipment Truck
DT	Display Truck
VCC	Virtual Control Center
CER	Communication Equipment Room
COTP	Communication On The Pause
FMX	Flexible Multiplexer
OSN	Optical Switching Network
UHF	Ultra-High Frequency
SHL	Switch Hot Line
M3SR	Multimode Multirole Multiband Surface Radios
DTS	Discipline Timing System
IFF	Identification Friend or Foe
ECCM	Electronic Counter-Counter Measure

Dte.	Directorate
NWC	Network Command
NWB	Network Bases
AE&S	Air Field Electronics and Simulator
ADAS	Air Defence Auto Data System
AVCS	Avionics
SRR	Short Range Radars
SAMS	Surface to Air Missile Systems
ADAS	Air Defence Automation System
Gnd Tpt	Ground Transport
RL	Radio Link
HCLOS	High Capacity Line Of Sight
RCN	Report Control Number
EMI	Equipment Maintenance Inspection
LP	Local Purchase
LR	Local Repair
ROCP	Radar Out of Commission for Parts
AROCP	Anticipated ROCP
UAL	Unit Authorization List
SOB/ SIB	Store Outward Book/ Store inward Book
MT	Mechanical Transport

MTD	Mechanical Transport driver
CD	Civilian Driver
RRR	Rapid Runway Repair
CU	Common User
Splt Veh	Specialist Vehicle
Ops Veh	Operational Vehicle
MTO	Mechanical Transport Officer
BA NO	Broad Arrow Number
MTG	Mechanical Transport Greaser
TP	Tyre Pressure
RPM	Revolution Per Minute
VMS	Vehicle Management system
TPRMS	Tyre Pressure Rotation Management System
PSI	Pond Square Inch
RCN	Return Control Number
MPG	Miles Per Gallon
BF	Bring Forward
RVT	Radio Vehicle Transport
TOs	Technical Orders
POL	Petrol Oil Lubricants